



Drive comfort and safety evaluation for vortex-induced vibration of a suspension bridge based on monitoring data

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ABSTRACT

Vortex-induced vibration (VIV) caused by wind excitation can seriously affect traffic comfort and safety. Therefore, reasonable control measures in the initial stage of VIV shall be formulated by the bridge management department to ensure the safety of passing vehicles. In this paper monitoring data of VIV of a suspension bridge from 2013 to 2017 were analyzed and the characteristics of structural vibration response factors such as wind parameters and Energy Concentration Coefficient (ECC) of VIV were studied. A dynamic monitoring model which used for VIV prediction was established based on the massive data collected by the existing structural health monitoring system and dynamic monitoring system of VIV installed on the bridge which is located in the East China Sea. The data measured in 2017 by the anemometers and acceleration sensors from dynamic monitoring system was taken as input to simulate the prediction process of VIV. The warning mechanism of VIV with 5 levels and corresponding measures were proposed. Results show that the root mean square (RMS) value of acceleration and ECC can be used as further criteria for judging the occurrence of VIV. The comprehensive decision method and trend analysis method can be used for VIV judgment. The recognition rate of the occurrence VIV is 88.00% and 93.24% by the comprehensive decision method and trend analysis method respectively, based on the dynamic monitoring model. The dynamic monitoring model for VIV can be adopted to timely predict VIV, and can be effectively applied in practical engineering.

1. Introduction

Since the beginning of the 21st century, modern bridges are developing towards large and super span. With the increase of bridge span, as well as the characteristics of lower damping, thin and flat box girder, light-weight, wind-induced vibration problems become more and more prominent. One of them is vortex-induced vibration (Hansen et al., 2015; Xu, 2013), a resonant and self-excited type wind-induced vibration in long-span bridges under lower wind speed. Therefore, its occurrence frequency is higher, which may cause structural fatigue damage in steel component in severe cases, and also affect the comfort and safety of driving (Zhu et al., 2018; Li et al., 2016). Examples of full-scale vibrations originating from vortex shedding are Great Belt Suspension Bridge in Denmark just before completion (Larsen et al., 1999), Osteroy Suspension Bridge in Norway (Isaksen, 2001) and Trans-Tokyo bay crossing bridge in Japan (Fujino and Yoshida, 2002).

As the wind stability is guaranteed for most long span bridges today,

more and more attention has been paid to the vortex-induced vibration and a series of studies have been carried out, which have effectively solved some problems in engineering practice and provided technical guidance for engineering practice. There are mainly research methods (Xu et al., 2010) for vortex-induced vibration of long-span Bridges, such as field measurement method, semi-empirical method, wind tunnel test and CFD numerical simulation.

The field measurement method is to measure the VIV response of the actual wind environment, structure and components in the field by using anemometer, accelerometer, displacement meter, data acquisition system. It is considered to be the most ideal method to study VIV. It can truly reflect the VIV of bridge structures in the natural wind environment. The data of wind parameters (Xu et al., 2019), the VIV response of structure (Li et al., 2017, 2018) under the natural wind environment are collected by the sensors such as the anemometer, accelerometer and displacement meter. Also it can obtain detailed and comprehensive wind field characteristics and VIV response data of components by using data acquisition system.

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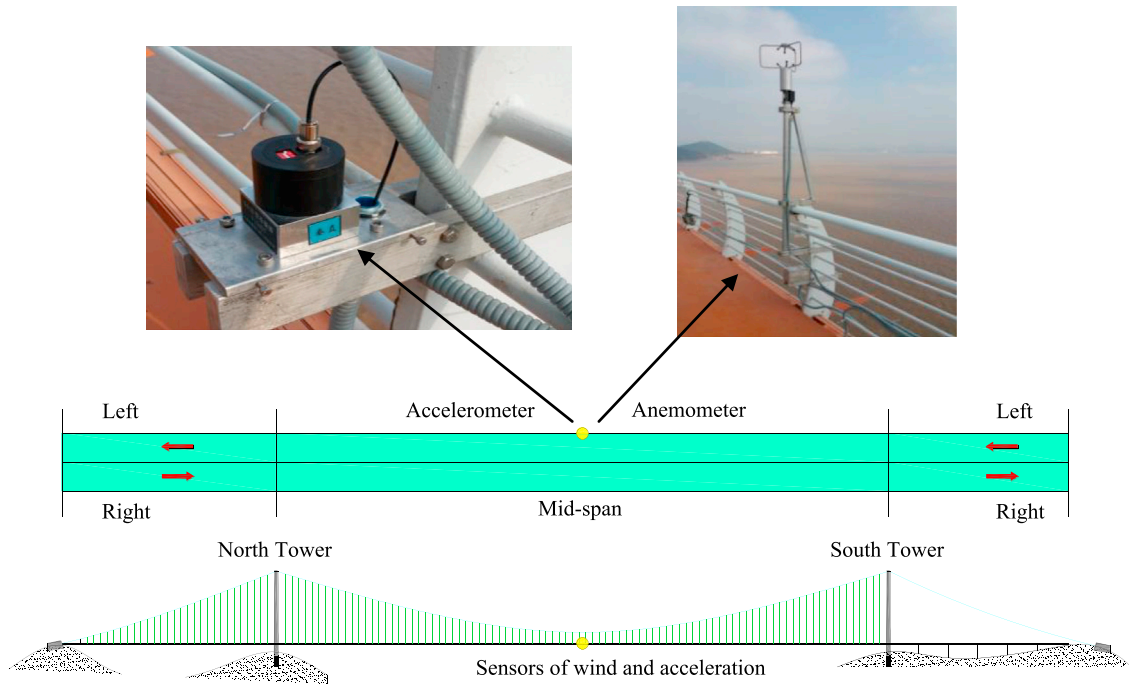


Fig. 1. The layout of partial sensor installed on the bridge of east China sea.

Table 1
Parameters of main equipment.

Device name	Photo of entity	Parameters
1. Data Acquisition Instrument		MX1615B(Germany) HBM Number of channels:16 Max sampling rate:19.2 kHz/channel Acquisition accuracy:24 bit
2. Three-dimensional ultrasonic anemometers		Wind speed:0–40 m/s, accuracy: ± 0.05 m/s Wind direction: 0–360° Wind attack angle: $\pm 60^\circ$, accuracy: $\pm 2^\circ$
3. One-direction acceleration sensor		frequency range: 0.17–100 Hz Maximum range: 20 m/s ² Sensitivity: 0.3 V/(m/s ²)

Semi-empirical method which is based on the theory of structural vibration is the most basic method to study the VIV. The differential equation of structural vibration is established to obtain the wind load acting on the structure and dynamic response. Some aerodynamic parameters, such as flutter derivatives, in the equation also are obtained through experiments or wind tunnel test. The analytical model (Ehsan. F, 1990; Larsen. A, 1995), wake-oscillator models (Hartlen and Currie, 1970), force-coefficient data models (Iwan and Blevins, 1974), scale

models (Nguyen D.T, 2018), etc, were simulated the bridge models (Marra et al., 2015) when VIV occurs.

Wind tunnel test is usually to measure the vibration response of scale segment model specimen in uniform flow field in the laboratory. A large number of wind tunnel tests show that turbulence has an obvious effect on suppressing vortex induced vibration of bridge structure. The effects of turbulence scale on vortex induced vibration have been some exploratory results (Daniels S. J. 2016). When the mechanism of vibration and the calculation method are not clear, wind tunnel test (Xu et al., 2009) is the most direct and effective method to predict the response of the bridge in natural wind field. The first attempt to model VIV of bridge decks was made by Scanlan (Marra et al., 2015). The aerodynamic response of bridge model with a similarity ratio was obtained in wind tunnel test (Li et al., 2017; Hua et al., 2015).

Because of the high cost of wind tunnel test, the numerical simulation method has developed rapidly in recent years. Based on the principle of aerodynamics and computational fluid dynamics technology (Zhu et al., 2007; Shuji and Ueda, 2003), the CFD numerical simulation method which relies on high-performance computers can simulate the interaction between wind and structure (Lucor. D, 2001; Li et al., 2017; Zhu, L.D., 2017). Some calculation parameters are taken from field measurement or test. The CFD numerical simulation is not enough accurate for predicting aeroelastic phenomena. At present, CFD cannot independently carry out research on vortex induced resonance of long-span bridges without wind tunnel test.

Table 2
Information about typical VIV events from 2013 to 2016.

Serial number	Frequency (Hz)	Modal order	Wind speed (m/s)		Wind angle (degree)		maximum RMS of acceleration (cm/s ²)	Start time	Duration (min)
1	0.3274	6	7.82	9.18	137	146	13.36	2013-04-10 19:10	110
2	0.3261	6	8.81	10.56	124	134	39.22	2013-07-11 17:50	70
3	0.3297	6	10.731	11.535	132	138	6.608	2014-02-08 09:40	40
4	0.3251	6	4.336	4.833	301	304	29.678	2014-06-07 15:10	160
5	0.3261	6	8.389	8.389	308	308	4.847	2014-09-12 15:10	10
6	0.3257	6	9.445	10.972	128	142	34.333	2014-10-14 09:10	300
7	0.3284	6	10.45	10.72	311	315	7.02	2015-01-06 06:40	30
8	0.3274	6	8.97	9.68	134	138	9.97	2015-01-24 14:00	40
9	0.3261	6	9.3	10	132	136	16.17	2016-06-17 16:30	60
10	0.3254	6	7.8	9.6	128	136	25.59	2016-07-25 12:40	260

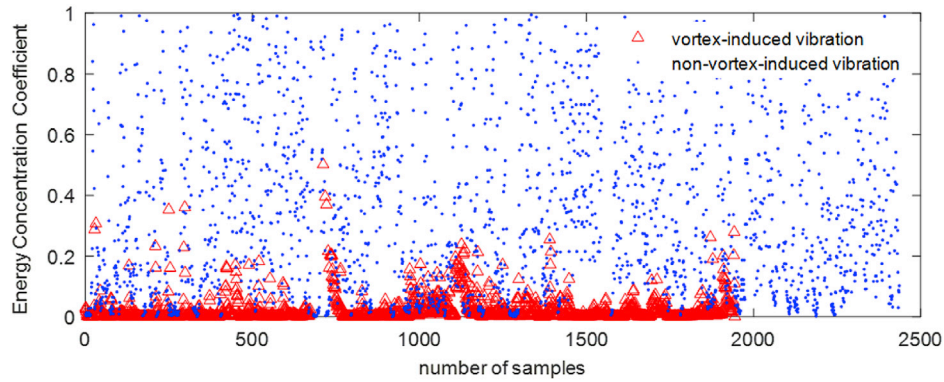


Fig. 2. ECC of VIV and non-vortex-induced vibration.

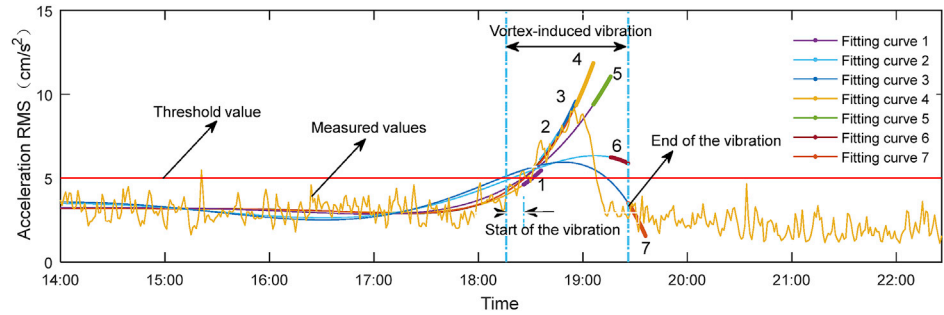


Fig. 3. Measured acceleration RMS and fitting curve.

Table 3
The corresponding relation between K_D and the comfort level of pedestrians.

Dynamically-sensitive parameter K_D (mm/s^2), (UK law BD 49/01)	Amplitude of acceleration A_D (m/s^2)	Characterization of human body
1	0.04	slight feel
2	0.08	acceptable
3	0.12	tolerable
5	0.20	
10	0.39	unpleasant
20	0.78	
30	1.18	
50	1.95	uncomfortable
100	3.90	

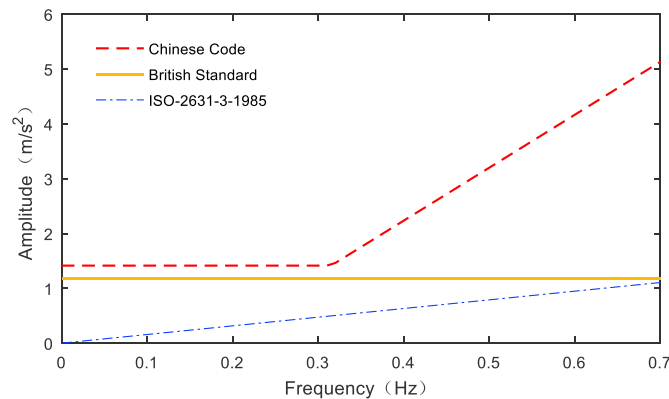


Fig. 4. Limits of VIV in different specifications.

Now, many long-span suspension bridges have equipped with structural health monitoring system (SHMS), including anemometers and vibration sensors, which provides a good basis for the field measurement study of wind-induced vibration response. It becomes a very important subject in bridge wind engineering field to study and analyze these measured data.

Previous researches of VIV are focused on influencing factors, experimental verification and occurrence mechanism, and control measures. Most of the research results are used in bridge design and safety assessment, but there are few studies on the occurrence or impending of vibration events. Some specific characteristic rules of vortex-induced vibration monitoring data are analyzed. The non-uniformity of down-wind wind speed had a significant impact on the vortex-induced vibration mode, and that wind speed, wind direction and non-uniformity were key parameters of the entire process and response amplitude of vortex-induced vibration (Li et al., 2017). Vortex-induced vibration was more likely to occur in the range of low wind speed, and the wind direction was almost perpendicular to the bridge line, with low turbulence intensity (Li et al., 2011, 2014). But further studies on the comprehensive judgment method, occurrence and development trend analysis and early warning mechanism of VIV based on these rules are still lacking.

The dynamic monitoring, prediction and traffic safety evaluation of vortex induced vibration has been studied by us. In the last article (Xu et al., 2019), the investigated bridge and corresponding health monitoring system are introduced, the wind and vibration characteristics at the bridge site are analyzed in detail, the major influencing factors were identified and the VIV identification model is established. In this paper, the dynamic monitoring system is established and the method of judgment of VIV is discussed extendedly. Warning mechanism of VIV is proposed. Evaluation index of VIV for pedestrian comfort, driving comfort and safety are analyzed. The dynamic monitoring model is presented and applied based on major influencing factors.



Fig. 5. Interference of the line of sight by vortex-induced vibration.

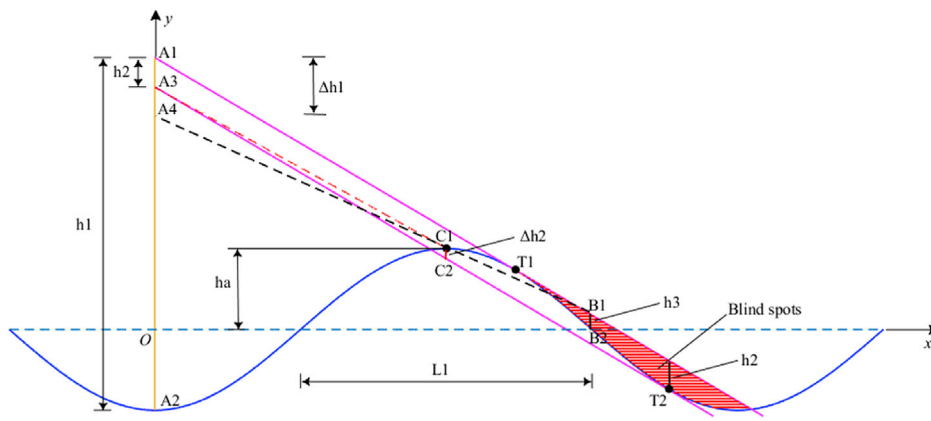


Fig. 6. The geometric model of the most unfavorable situation of the line of sight.

2. Description of the dynamic monitoring system

A full set of structural health monitoring system was installed on the suspension bridge with a main span of 1650 m during the construction stage, including environmental temperature, wind speed, wind direction, strain, vibration acceleration, displacement, and other structural response parameters. The accumulated monitoring data of past decade provide effective research foundation for the research of this paper. The existing structural health monitoring system including the layout of anemometer and vibration sensor was described in detail (Xu et al., 2019). The wind field and vibration acceleration data monitored from 2014 to 2017 are selected to perform factors analysis of vortex-induced vibration.

In addition, an independent dynamic monitoring system was installed especially for dynamic monitoring and prediction of VIV. The layout of the anemometers and acceleration sensors of dynamic monitoring system are shown in Fig. 1. The dynamic monitoring system of VIV is divided into four subsystems, namely, sensor subsystem, data acquisition and transmission subsystem, data storage subsystem, data processing and analysis subsystem. The parameters of hardware equipment of sensor subsystem are shown in Table 1. Through the comparative analysis, it can see that the data collected by the two systems are basically the same. The independent dynamic monitoring system can capture all the VIV events, even if the VIV occurs at other vertical vibration mode. The data collected by dynamic monitoring system are mainly used for prediction of VIV. At the same time, it provides the basis for establishing the dynamic monitoring model of VIV.

3. Judgment of vortex-induced vibration

3.1. Influencing factors analysis of VIV

The occurrence of VIV should be closely related to the characteristics of wind field and vibration. It is necessary to study the characteristics of wind parameters including average wind speed, wind direction, turbulence intensity, gust factor, and vibration parameters including RMS of acceleration and ECC during the occurrence of VIV. The above parameters and its distribution rules were deeply analyzed based on the data collected from 2013 to 2016 (Xu et al., 2019). It provides a reliable data base for the establishment of dynamic monitoring model. Some typical VIV events were shown in Table 2.

The ‘frequency lock-in’ is common in VIV and the wind speed range in which frequency lock-in occurs is called the ‘lock-in wind speed range’. According to the occurring frequency of wind speed, the lower order vertical bending mode is more likely to appear. The FE model of bridge was established and the frequency of the first to eighth vertical vibration modes of the bridge was 0.095 Hz, 0.133 Hz, 0.181 Hz, 0.231 Hz, 0.266 Hz, 0.327 Hz, 0.434 Hz and 0.435 Hz. It can be seen that the sixth mode is the one that resonates in this VIV event. The sixth mode is the best combination of wind energy and low turbulence, so VIV is more likely to occur at this mode. The range of wind speed and wind angle is consistent with the content of Section 3.2.

The ECC is defined as the ratio of power spectral density and shown in equation (1).

$$Ce = W_{p2}/W_{p1} \quad (1)$$

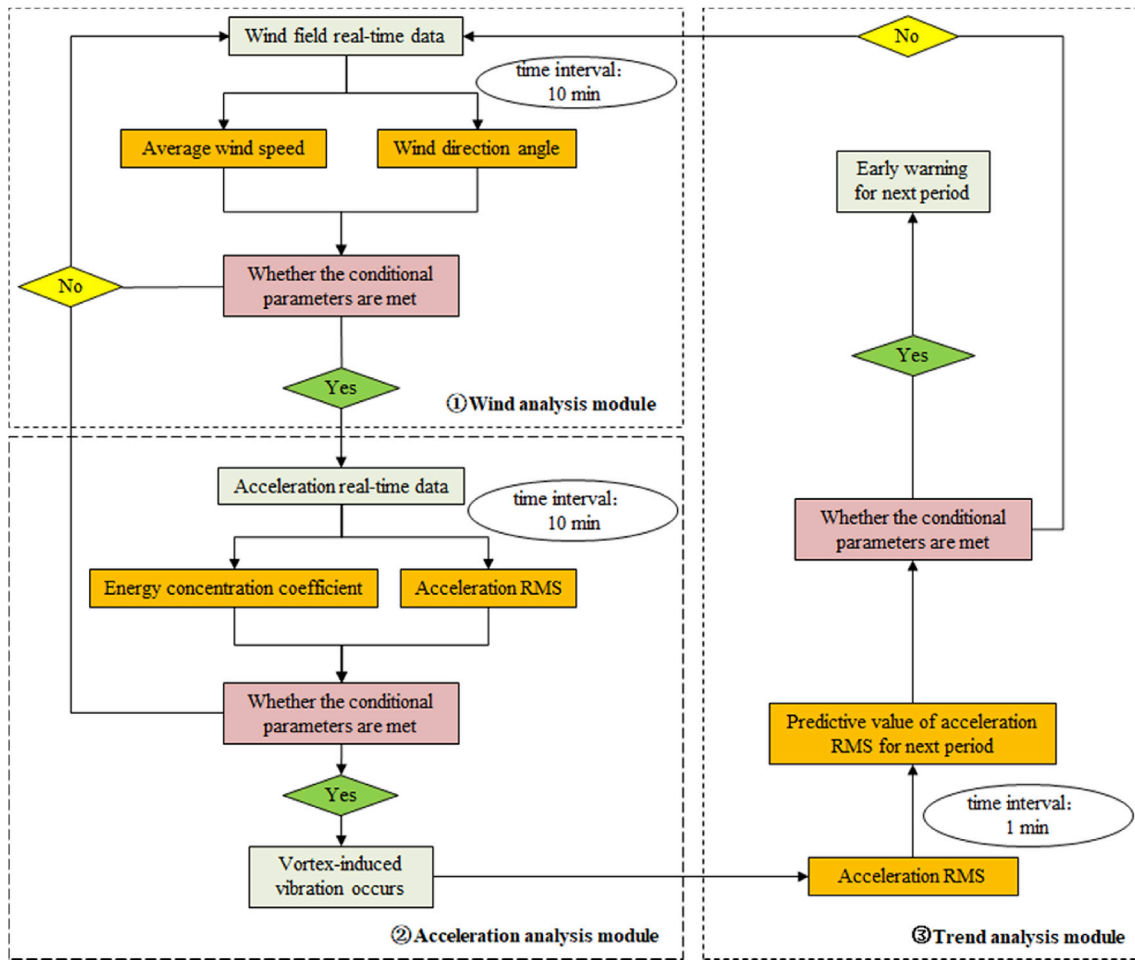


Fig. 7. Dynamic monitoring model framework of VIV.

Table 4

Warning mechanism of VIV for long-span suspension bridges.

K_D (mm/ s^2)	Warning level	Warning measures
<5	Level 5	Small amplitude, no warning
[5, 10)	Level 4	Pay close attention to the amplitude of mid-span
[10, 15)	Level 3	Issue early warning information, careful driving
[15, 30)	Level 2	Large amplitude, it is suggested to interrupt the traffic and forbid vehicles and pedestrians to pass
≥ 30	Level 1	Not allowed

where, C_e is ECC, W_{p1} is the maximum value of power spectral density in the spectrum analysis of the main girder and W_{p2} is the second-highest value of power spectral density.

The analysis results of ECC from 2013 to 2016 and partial non-vortex-induced vibration data are extracted and compared which is shown in Fig. 2. The number of samples of VIV and non-VIV are 1944 and 2436, respectively. We determine the threshold of ECC by 0.9 segment position, that is $ECC \approx 0.1$. So, the percentage of the non-vortex-induced vibration data above and below the threshold is 73.52% and 26.48%, respectively. There are 91.36% and 8.64% in VIV events. It can be seen that most of ECC in the non-vortex-induced vibration process are between 0.1 and 1, while it was generally between 0 and 0.1 during the VIV. When VIV occurs, the ECC value of some samples is greater than 0.1. This is because, on one hand, the time period of VIV is not exactly selected, and the beginning and end segments may contain some non-VIV data; on the other hand, when multiple VIV occurs continuously, the middle segment can contain some non-VIV data. Therefore, the distribution rule of ECC

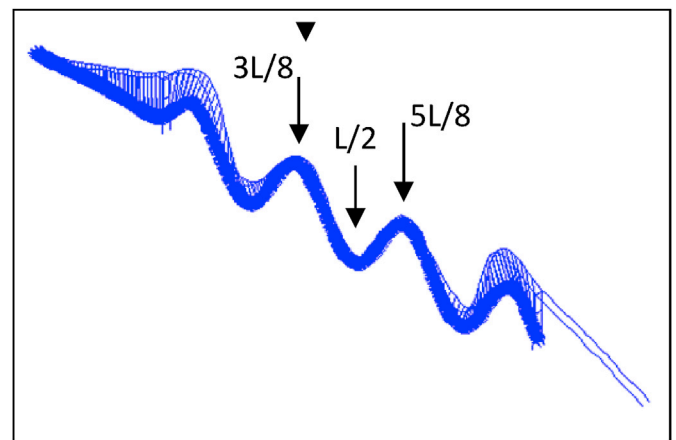


Fig. 8. The sixth vibration mode of the bridge.

can be used as an important basis for the early warning of VIV.

3.2. Comprehensive decision method

Through the above analysis, four parameters, including average wind speed, wind direction, RMS, and ECC, are chosen as the parameters to estimate the occurrence of VIV. A comprehensive decision method for the occurrence of VIV is established based on the four parameters and mainly divided into two parts. One is the wind speed analysis module based on

Table 5

Recognition result of dynamic monitoring model with an analysis interval of 10 min.

Serial number	Data analysis report of the structural monitoring system			Dynamic monitoring model		
	Starting time	Ending time	Duration	Recognition duration	Recognition results	Level
1	2017-01-22 01:10	2017-01-22 01:30	20	20	Yes	5
2	2017-01-22 02:00	2017-01-22 02:40	40	40	Yes	5
3	2017-02-15 13:30	2017-02-15 13:40	10	10	Yes	5
4	2017-02-15 14:30	2017-02-15 14:40	10	10	Yes	5
5	2017-03-03 13:30	2017-03-03 13:50	20	20	Yes	5
6	2017-03-03 18:00	2017-03-03 19:00	60	50	Yes	5
7	2017-04-13 19:00	2017-04-13 19:40	40	30	Yes	5
8	2017-05-19 17:30	2017-05-19 17:50	20	20	Yes	5
9	2017-05-19 18:20	2017-05-19 20:10	110	110	Yes	5
10	2017-05-20 16:00	2017-05-20 16:50	50	50	Yes	5
11	2017-05-20 21:10	2017-05-20 21:20	10	/	No	/
12	2017-05-30 11:10	2017-05-30 14:00	170	180	Yes	3
13	2017-07-15 14:50	2017-07-15 15:50	60	30	Yes	4
14	2017-07-15 16:50	2017-07-15 18:10	80	120	Yes	3
15	2017-07-16 13:30	2017-07-16 14:10	40	50	Yes	5
16	2017-07-16 16:30	2017-07-16 16:50	20	/	No	/
17	2017-07-18 01:50	2017-07-18 02:30	40	50	Yes	5
18	2017-08-03 15:00	2017-08-03 16:00	60	60	Yes	5
19	2017-09-05 19:00	2017-09-05 21:10	130	100	Yes	4
20	2017-09-17 02:00	2017-09-17 02:30	30	30	Yes	5
21	2017-09-17 07:00	2017-09-17 10:30	210	210	Yes	4
22	2017-09-19 06:20	2017-09-19 07:00	40	40	Yes	5
23	2017-09-19 17:50	2017-09-19 18:10	20	/	No	/
24	2017-09-19 19:10	2017-09-19 22:00	170	170	Yes	3
25	2017-09-27 13:00	2017-09-27 14:30	90	80	Yes	5
The accuracy of recognition				22/25 = 88.00%		

the preliminary judgment of real-time monitoring data of wind speed. The other is the acceleration analysis module based on real-time monitoring data of acceleration. Through the analysis of monitoring data, the four characteristic parameters shall meet the following conditions respectively when VIV occurs:

- The wind speed ranges from 5 to 13 m/s.
- The wind direction is distributed in the range of 120°-150° and 300°-330°.
- The ECC decreases approximately 0 in a short time and is less than 0.1.
- The RMS of acceleration increases quickly in a short time and is greater than 5 cm/s².

It is assumed that the number of anemometers and acceleration sensors are m and n respectively. α_i, β_i ($i = 1, 2, \dots, m$) and μ_j, ν_j ($j = 1, 2, \dots, n$) represent the status of whether the conditions of the parameters are met, with 1 and 0 standing for condition satisfied or not. For example, $\alpha_i = 1$ or $\beta_i = 1$ means the average wind speed or wind direction meets the threshold value of VIV, while $\mu_j = 1$ or $\nu_j = 1$ means the RMS value or ECC meets the threshold value of VIV.

The number of conditional parameters is $2m$ and $2n$ respectively. In the process of VIV, it is less likely that all conditional parameters are met at the same time, usually only a part of the conditions are satisfied. If the number of wind field parameters and vibration characteristics parameters that meets the conditions is p and q respectively, equation (2) can be used as the judgment of the occurrence of VIV. Through multi-objective optimization and trial calculation, the values of p and q are defined as m , $n+1$ respectively. Based on the $p = m$ and $q = n+1$, the prediction model (see in Fig. 7) can obtain high recognition rate and low misrecognition rate (Xu et al., 2019).

$$\begin{aligned} \sum_{i=1}^m \alpha_i + \sum_{i=1}^m \beta_i &\geq p, \quad (i = 1, 2, \dots, m) \\ \sum_{j=1}^n \mu_j + \sum_{j=1}^n \nu_j &\geq q, \quad (j = 1, 2, \dots, n) \end{aligned} \quad (2)$$

3.3. Trend analysis method

Four parameters, including the average wind speed, the wind direction, the RMS value, and the ECC, are chosen as the indexes to determine the occurrence of VIV. Because the RMS value can more fully express the varying intensity of the vibration acceleration, the RMS value of the vibration acceleration data was calculated (Xu et al., 2019). In the course of VIV, the RMS value of main bridge vibration is almost above 5 cm/s²; and most of the RMS value is below 5 cm/s² in normal vibration. Therefore, the value of 5 cm/s² is taken as the VIV threshold.

During the process of VIV, there is a crest given by the RMS value. The change trend of VIV in the coming period can be reflected by using this feature of acceleration RMS. The measured acceleration RMS and its fitting curve at each analysis period of a typical VIV event that lasted about 70 min from PM18:16 to PM19:26 were shown in Fig. 3.

In the trend analysis module, the time interval is set as 1 min for RMS. The 10-min time interval was used to the predictive value of acceleration RMS for next period. It determines whether VIV occurs according to the predictive value of acceleration RMS. If the predictive value of RMS (in 10-min time interval) large than 5 cm/s², the VIV occurs. For each analysis interval, acceleration RMS in the previous period was fitted to obtain the fitting value of the next analysis interval. If its fitting value exceeds the threshold value of RMS condition parameter, it is predicted that VIV will occur in the next analysis interval; otherwise, VIV will not occur. And so on, until the end of this VIV event. For example, the fitting curve 1 represents the fitting prediction results in the analysis interval from PM18:16 to PM18:26, and the fitting value exceeds the threshold, indicating that VIV will occur in the next analysis interval from PM18:26 to PM18:36. The fitting curve 7 represents the fitting prediction results in the analysis interval from PM19:16 to PM19:26, and its fitting value does not exceed the threshold, indicating that VIV will not occur in the next analysis interval from PM19:26 to PM19:36, and that is the end of this VIV event. The fitted curve from the beginning of vibration to the end better reflects the development trend of the VIV over time. Therefore, the acceleration RMS value can be used as the characteristic parameter of the trend analysis for VIV.

Table 6

Results of trend analysis with an analysis interval of 10 min.

Serial number	Starting time	Duration	N	N_e	The accuracy of trend analysis
1	2017-01-22 01:10	20	2	2	100.00%
2	2017-01-22 02:00	40	4	4	100.00%
3	2017-02-15 13:30	10	1	0	0.00%
4	2017-02-15 14:30	10	1	0	0.00%
5	2017-03-03 13:30	20	2	2	100.00%
6	2017-03-03 18:00	50	5	4	80.00%
7	2017-04-13 19:00	30	3	3	100.00%
8	2017-05-19 17:30	20	2	1	50.00%
9	2017-05-19 18:20	110	11	11	100.00%
10	2017-05-20 16:00	50	5	5	100.00%
11	2017-05-20 21:10	10	/	/	0.00%
12	2017-05-30 11:10	180	18	18	100.00%
13	2017-07-15 14:50	30	3	2	66.67%
14	2017-07-15 16:50	120	12	11	91.67%
15	2017-07-16 13:30	50	5	5	100.00%
16	2017-07-16 16:30	20	/	/	0.00%
17	2017-07-18 01:50	50	5	5	100.00%
18	2017-08-03 15:00	60	6	6	100.00%
19	2017-09-05 19:00	100	10	9	90.00%
20	2017-09-17 02:00	30	3	3	100.00%
21	2017-09-17 07:00	210	21	21	100.00%
22	2017-09-19 06:20	40	4	3	75.00%
23	2017-09-19 17:50	20	/	/	0.00%
24	2017-09-19 19:10	170	17	17	100.00%
25	2017-09-27 13:00	80	8	6	75.00%
Total			148	138	93.24%

4. Warning mechanism of vortex-induced vibration

4.1. Evaluation index of VIV for pedestrian comfort

At present, the Chinese code (JTG/T 3360-01-2018), Japanese guide for short (Sato H, 2003) and UK law (BD 49/01) directly stipulate the vorticity limit of main girder in the bridge design codes at home and abroad. In addition, in the technical report of RWDI company, the vortex limit value of main girder is introduced that commonly used in North America and Britain (IRWIN P A, 1999). The Chinese code adopts the same vortex-vibration limit value h_a as the Japanese guide and the h_a is given as follows.

$$h_a = \frac{0.04}{f_b} \quad (3)$$

where h_a is the allowable amplitude of the vortex-induced vibration, f_b is the frequency of the VIV.

Since the VIV of a bridge generally presents simple harmonic vibration of a single mode, the displacement response of the vertical vortex-vibration can be approximated by sine or cosine functions. The allowable value of the corresponding acceleration amplitude can be calculated as follows:

$$A_a = \frac{0.04}{f_b} \omega_b^2 = \frac{0.04}{f_b} (4\pi^2 f_b^2) = 1.58 f_b \quad (4)$$

where A_a is the allowable acceleration amplitude of the vortex-induced vibration, ω_b is the circular frequency of the vortex-induced vibration.

BD 49/01 Design Rules for Aerodynamic Effects on Bridges used dynamically-sensitive parameter to evaluate the load effect of the VIV and its influence on pedestrian comfort as follows:

$$K_D = y_m f_b^2 \quad (5)$$

$$A_D = y_m \omega_b^2 = y_m f_b^2 4\pi^2 = 4\pi^2 K_D \quad (6)$$

where y_m is the amplitude of the VIV, K_D is the dynamically-sensitive parameter, and A_D is the acceleration amplitude of the VIV.

The corresponding relation between K_D and the comfort level of pedestrians, the conversion relation between K_D and the acceleration amplitude A_D are shown in Table 3. When K_D is greater than 30 mm/s², it may lead to the discomfort of pedestrians. The VIV modes of the investigated bridge are mainly of the sixth order, about 0.325 Hz, and its corresponding K_D is about 13 mm/s² according to the Chinese code.

4.2. Evaluation index of VIV for driving comfort

At present, ISO2631 is an international standard widely used for driving comfort evaluation. The driving comfort of automobile is mainly influenced by the vibration and impact environment on the occupant's comfort, so the driving comfort is mainly evaluated according to the occupant's subjective feeling of comfort. ISO-2631-3-1985 gives the root-mean-square limit of human vibration acceleration that may lead to motion sickness, which is called "extreme discomfort limit" and related to the vibration time and frequency suffered by human body. The vibration time was divided into three grades, namely 30 min, 2 h and 8 h. In the actual situation, the time the vehicle may stay on the bridge is usually less than 30 min, so the "30 min motion sickness limit" can be conservatively selected as the bridge vortex-vibration limit considering the influence of vehicle occupant comfort.

By comparing the acceleration amplitude limits corresponding to the Chinese code, the British law and the "30 min motion disease limit", as shown in Fig. 4, it can be found that both the Chinese code and the British rules are stricter than the "30 min motion disease limit".

4.3. Evaluation index of VIV for driving safety

Fig. 5 shows two pictures of different time recorded at observation points near the bridge tower when vortex-induced vibration occurs. The white electric box is located near the middle span of the bridge, and its vertical relative position at the same observation point changes obviously. Therefore, when the amplitude reaches a certain degree, the waveform curve may interfere with the driver's line of sight, thus posing a threat to driving safety.

When the acceleration of bridge structure increases at low wind speed and contains only one frequency, it can be judged whether VIV occurs or not. Therefore, the vertical vortex-induced vibration mode of the bridge can be considered as simple harmonic. Assuming that the mode of vortex-induced vibration is simple harmonic, the calculation model takes into account the most unfavorable case of influence on the driving line of sight, namely the driving vehicle is located at the trough of the mode curve. At this time, no matter how many half-waves the vibration mode contains, the positions where the line of sight is most disturbed are all

located in the one-half wave range in front of the adjacent wave peak in the driving direction.

Chen and Huang (2015) proposed the geometric model of the most unfavorable situation of the line of sight, as shown in Fig. 6. As can be seen in Fig. 6, the h_a is the amplitude of the vortex-induced vibration, h_1 is the height of driver's line of sight and L_1 is the length of half-wave. There is a visual blind area in the closed area formed by tangent A_1T_1 and mode curve. Assuming that the height of the blind area at the position of point B_2 is h_3 , and the maximum invisible height in the blind area is h_2 , there is a geometric relationship between them: $h_2 = 2h_3$. According to the geometric model, the amplitude limits of vortexes can be approximately calculated when the maximum invisible height of the visual blind area is larger than a certain height. Chen and Huang (2015) used the auxiliary line A_4B_1 to approximately calculate the h_a . If the driver's line of sight h_1 is 1.2 m and the height of the obstacle h_2 is 0.1 m, then the approximate solution of the amplitude limit of vortexes is 0.2763 m.

4.4. Warning mechanism

The allowable amplitude h_a in Eq. (3) is mainly used for bridges with a span under two hundred meters, and it is conservative on long-span bridges (Sato H, 2003). It can only be used as a basis for a certain level of classification. Therefore, the early warning mechanism of VIV for long-span suspension bridges is established mainly by referring to K_D in British rules. The warning mechanism is set as five warning levels, corresponding to different dynamic sensitive parameter and warning measures, as shown in Table 4. The acceptable range is defined as level 5, the transition period is defined as level 4, the Chinese code limit is defined as level 3, the range between the Chinese code limit and the British rules limit is defined as level 2, and the range beyond the British rules limit is defined as level 1.

The acceleration amplitude of the bridge is employed to make 5 different warning levels. Also, the driving comfort (Ma et al., 2018) is directly related to the acceleration responses experienced by the driver in the vehicles. It's more reasonable to consider the full vibration responses of the driver when determining the warning levels for the purpose of drive comfort and safety evaluation. When VIV occurs, the vehicle-bridge coupling system is more manifested as the vehicle attached to the bridge up and down in the process of moving forward. This is different from the wind-vehicle-bridge resonance under buffeting. Although vehicle vibration (Ma et al., 2017) can regulate the transmission of bridge vibration, the frequency of vortex induced resonance is usually lower than 1.0 Hz. However, the vehicle frequency is much higher than 1.0 Hz and the vehicle vibration has limited effect on the acceleration regulation of VIV. Therefore, it can be approximately considered that the vertical vibration of the bridge is the vibration of vehicle drivers and passengers. Finally, the wind-bridge coupled dynamic analysis during VIV event is considered in this paper.

5. Application of the model

5.1. Dynamic monitoring model

According to the comprehensive decision method, the trend analysis method and warning mechanism, the dynamic monitoring model framework is established as shown in Fig. 7. According to the judgment of the wind speed and acceleration analysis module, when VIV occurs, the early warning information of VIV will be released. Meanwhile, it will enter the trend analysis module to predict the RMS value of the next analysis period. If the predicted value reaches the threshold of the conditional parameters for the occurrence of VIV, it can be considered that VIV will occur in the next analysis time interval.

In the first step (wind analysis module), the time interval is set as 10 min for wind speed and wind angle. Usually, the time interval is set as 10 min according to the Chinese bridge's wind engineering specification

(Wind-resistant Design Specification for Highway Bridge JTG/T 3360-01-2018).

In the second step (acceleration analysis module), the time interval is set as 10 min for ECC and RMS. The analysis results of this step (ECC and RMS) are used in the comprehensive decision method. It determines whether VIV occurs according to the aggregate score.

In the third step (trend analysis module), the time interval is set as 1 min for RMS. The 10-min time interval was used to the predictive value of acceleration RMS for next period. It determines whether VIV occurs according to the predictive value of acceleration RMS. If the predictive value of RMS (10-min time interval) large than 5 cm/s^2 , the VIV occurs.

5.2. Dynamic monitoring system of VIV

Due to the timeliness of VIV monitoring, the storage frequency of data such as acceleration and wind speed in the original monitoring system could not meet the requirements of real-time early warning of VIV. Some sensors even exceeded the range, resulting in data truncation. At the same time, the natural vibration response of the bridge is a random vibration response under the excitation of environmental conditions. Monitoring requirements and bridge structure form should be taken into consideration in determining the layout of measuring points. Acceleration sensors are generally arranged in the position with obvious vibration according to the vibration mode, preferably the crest or trough of the wave. Considering the resonant frequency of VIV is mainly occurs in the sixth order (3.5 waves) as shown in Fig. 8, the reasonable choice of acceleration monitoring points should be $3/8$ span, $1/2$ span and $5/8$ span of the main span. Therefore, it is necessary to establish an independent dynamic monitoring system of VIV. In the new system, one three-dimensional ultrasonic anemometer was installed on the cross-section in $1/2$ span of the main span, and three one-way acceleration sensors are mounted in the $3/8$ span, $1/2$ span and $5/8$ span of the main span. The details of installation are shown in Fig. 1.

5.3. Application of the model

According to the data analysis report of the structural monitoring system of the investigated bridge, a total of 25 vortex-induced vibrations occurred in 2017. Compared with the data collected by the independent dynamic monitoring system, the recognition results of the dynamic monitoring model are shown in Table 5. It can be seen that the warning level of vortex-induced vibration of the bridge in 2017 is in the range of level 1 - level 3, which indicates that the amplitude of vortex-induced vibration is within a certain allowable range. Vortex-induced vibration was identified for 22 times, and the recognition accuracy was 88.00%. For a few vortex-induced vibrations with short duration, it may be necessary to adjust the time interval to improve the accuracy.

The accuracy of trend analysis indicates the consistency between the trend analysis and the judgment results of VIV. The calculation expression of a VIV event is shown as follow.

$$P = \frac{N_e}{N} \times 100\% \quad (7)$$

where P is the accuracy of trend analysis, N_e is the number of times the trend analysis is consistent with the judgment, and N is total times of trend analysis. The measured analysis results are shown in Table 6. For a single VIV event, the longer duration of VIV, the higher accuracy of trend analysis. The total accuracy of trend analysis is 93.24%. From the test results, the dynamic monitoring model can better reflect the development trend of VIV in the next analysis period, and can be used for dynamic monitoring of the entire VIV process.

6. Conclusions

According to the data collected by the existing structural health

monitoring system and dynamic monitoring system, the factors analysis, judgment method, warning mechanism and dynamic monitoring model of vortex-induced vibration have been studied, and the following conclusions could be drawn:

- (a) Vortex-induced vibration can be well judged by the comprehensive decision method and trend analysis method depends on characteristic parameters.
- (b) The warning mechanism with 5 levels and measures was proposed according to the dynamic sensitivity parameters K_D .
- (c) The dynamic monitoring model is established, and the results of the model application show that the recognition rate of VIV is 88.00% by the comprehensive decision method and 93.24% by the trend analysis method.

Monitoring data measured from 2013 to 2017 by anemometers and acceleration sensors are used for VIV prediction, a long time monitoring is still needed to improve prediction accuracy.

CRediT authorship contribution statement

Sugong Cao: Conceptualization, Methodology, Investigation, Visualization, Writing - original draft. **Yong Zhang:** Validation, Formal analysis, Visualization, Software. **Hao Tian:** Formal analysis, Visualization, Supervision. **Rujin Ma:** Writing - review & editing, Supervision. **Weijie Chang:** Resources, Data curation, Validation. **Airong Chen:** Writing - review & editing, Supervision.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

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